

- ① Design & Config. of Star, bus, Ring topologies using PT
- ② Design the functionalities & Exploration of Top using PT

O/P
Verified
✓

Answer

① Aim:- To implement a star, bus, Ring topologies using Packet tracer and hence to transmit data between the devices connected using topology.

Apparatus required:- Packet Tracer, end devices, bridge, connectors

Procedure:-

Step 1:- Open Packet Tracer

* create a new network project

Step 2:- create Star Topology

* Place one switch at the centre

* Place multiple PCs around the switch

* Assign IP Address to each PC, for exam

• PC0 -> 192.168.1.1

• PC1 -> 192.168.1.2

- PC2 $\rightarrow$ 192.168.1.3
- PC3 $\rightarrow$ 192.168.1.4

Aim: To
Function
Test

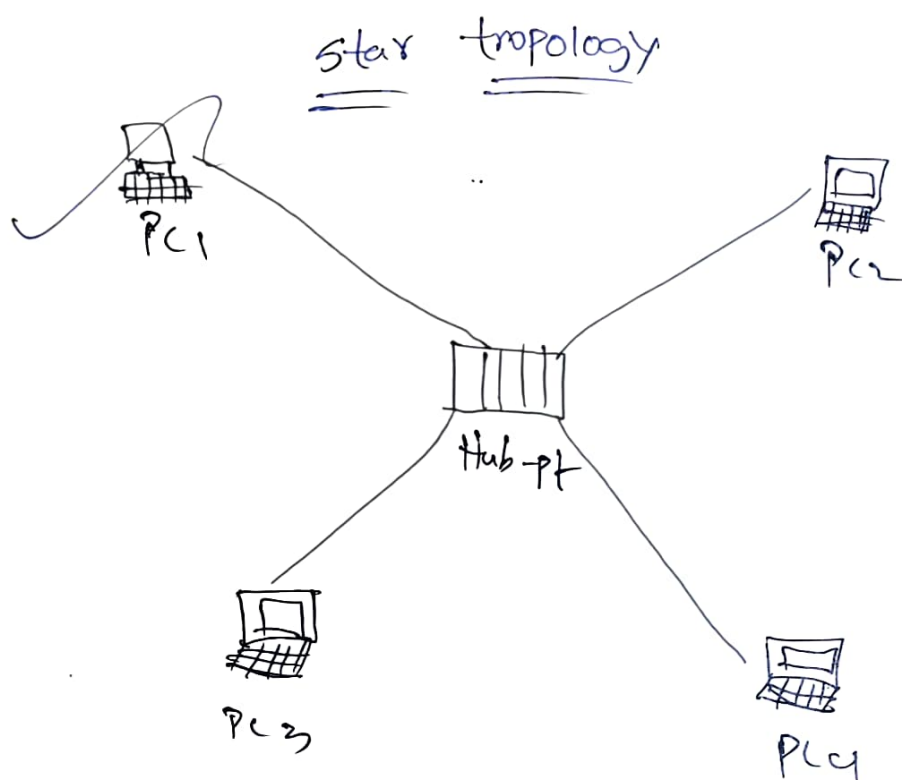
Step 3:- create bus topology

- * Place a hub in packet tracer to represent the shared communication medium.
- * connect the multiple PCs to the hub
- * connect the devices using ethernet cables.

Step 4:- create Ring Topology

- * place a multiple switches in packet tracer.
- * connect PCs to the switches.
- * Assign suitable IP addresses to the PCs.

Diagram:-



Aim:- To design the explore the various functionalities of cisco packet tracer and testing communication between devices.

Apparatus required:-

- * Cisco packet tracer
- * PC/End devices
- * Switches
- * Hub + Ethernet cables
- * Network interface ports

Procedure:-

Step 1:- * Open Packet Tracer

 * Create a new network project

* From the network devices section, select devices such as:-

- * Router, switch, hub

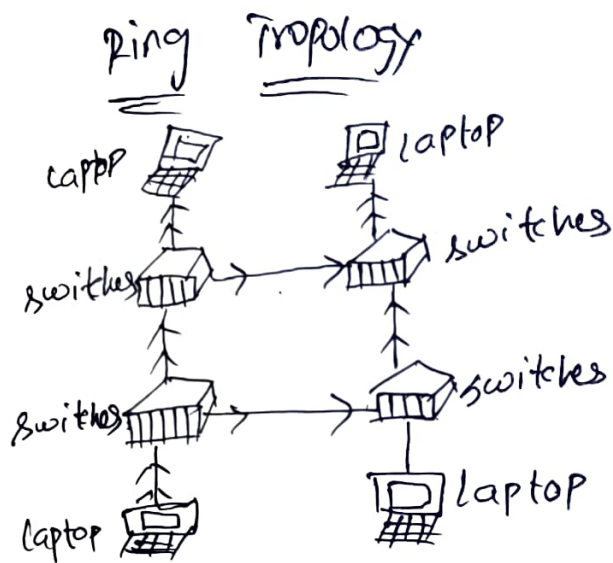
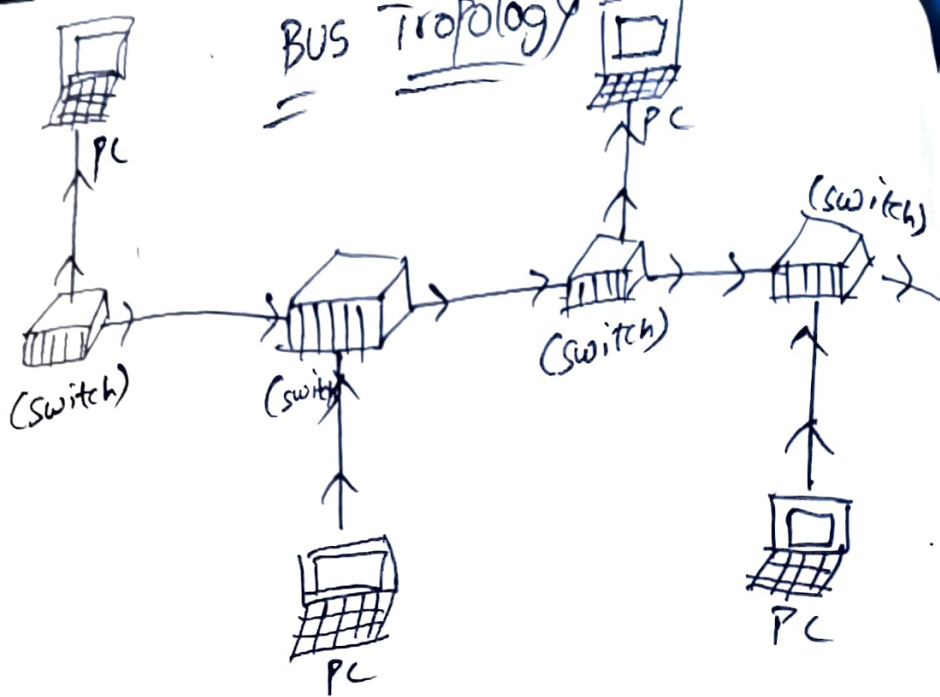
* From end devices, select PCs and place them in the workspace.

* Connect the IP addresses of each PC

* Observe whether the packets are successfully transmitted

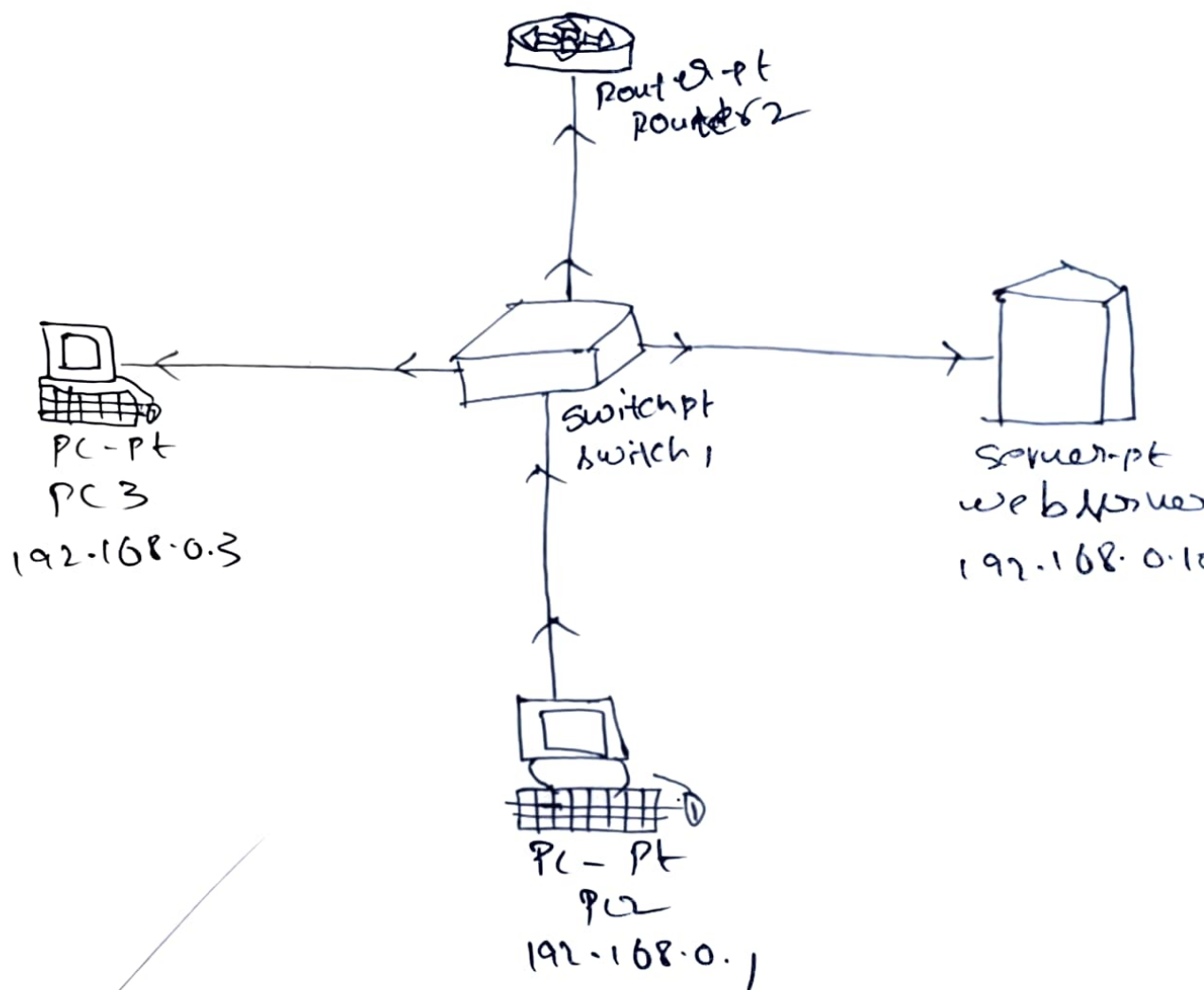
* Switch between Realtime mode and Simulation mode

* Explore the different topologies such as star, bus and ring and observe the working.



Result:- Thus, the star, bus and Ring network topologies were successfully designed and configured using Cisco Packet Tracer.

Diagram :



Result:- Thus, the functionalities of Cisco Packet Tracer were successfully explored. Network devices were added and connected using Cisco Packet Tracer.